

Technical Reference Document Listing Antimicrobial Agents of Veterinary Importance for Swine

An appendix to the List of Antimicrobial Agents of Veterinary Importance

Scope

The objective of this Technical Reference Document Listing Antimicrobial Agents of Veterinary importance for Swine (hereafter, referred as the “Technical Reference Document”), is to provide additional, species specific information without serving as a treatment guideline. By identifying antimicrobial agents used in swine, it can contribute to the development and update of national treatment guidelines, advice on veterinary medical use of veterinary medicinal products and best practice management, risk management, and risk prioritisation to minimise and contain antimicrobial resistance (AMR).

It should be kept in mind that the antimicrobials listed in this technical reference document may not all be available in all countries or be appropriate for use in all stages of the production cycle.

It is acknowledged that the situation is very diverse in different regions for licensing, availability, off-label use, and resistance to antimicrobial agents, and that the general information provided in this document should be interpreted in light of the local context. Antimicrobials used solely for non-medical purposes (i.e. growth promotion) or used off-label for treatment of certain infectious diseases in some geographical regions were not included in the document.

Swine-related recommendations stated in the World Organisation for Animal Health (WOAH) Standards and Guidelines (namely the [WOAH List of Antimicrobial Agents of Veterinary Importance](#), hereafter the “WOAH List”) should be considered alongside this document.

Methodology to prepare this document

A subgroup from the Antimicrobial Resistance Working Group (Appendix 1a) was nominated by WOAH to work on the development of the Swine Technical Reference Document. As a first step, an evidence-guided rapid review was undertaken by the Subgroup to prepare a preliminary table of important bacterial pathogens of swine animals and the antimicrobial agents used to treat these pathogens.

For the preparation of this table of swine pathogens, two globally focused reviews of swine diseases published in the last 20 years were consulted for swine pathogens and recommended treatments. The most detailed review was that contained within Diseases of Swine (Zimmerman et al. 2019) and to a lesser extent, within Pig Diseases (Taylor, 1999). To commence the project, a thorough review of the chapters devoted to bacterial diseases (Chapters 47 -64, pages 743-1002) was undertaken and a table of disease names, causative pathogens and treatment options was compiled.

To assess the completeness of the information extracted from Zimmerman et al. (2019), the relevant content in three contemporary guidance documents (the Australian Veterinary Association’s Antimicrobial prescribing for pigs 2020; the Swedish Veterinary Association’s Guidelines for the use of antimicrobial agents in production animals: cattle, pigs, sheep and goats from 2013 and the *Terrestrial Animal Health Code* 2021) were examined and new information integrated into the summary draft table of pathogens.

Additional sources of information used were:

- The original answers to a questionnaire sent to WOAH Members in 2006, which formed the basis for the current WOAH List. The answers to this questionnaire contain information on antimicrobials used to treat pathogens by animal species.
- List of antimicrobials authorised for the named species in countries

- Existing specific treatment guidelines
- OIE ad hoc Group report on vaccines that can reduce the use of antimicrobials

The end product of the review was a table presenting the following information:

- Disease;
- Pathogen involved;
- Antimicrobial class;
- Antimicrobial sub-class;
- Molecule;
- Comments and other considerations.

Once this table was established by the Subgroup, it was submitted to a panel of external swine health experts (Appendix 1b). These experts were identified through the WOAHA Collaborating Centres and the Working Group members and WOAHA networks. The experts represented geographical areas with sizeable swine populations and different areas of expertise in swine health management and industry. The experts were asked to address gaps in knowledge identified by the Subgroup and to provide feedback concerning the tables of antimicrobials, list of major pathogens and diseases and the proposed indications for use of antimicrobial groups.

The Subgroup took into consideration the feedback provided by the experts to consolidate the Technical Reference Document. The Subgroup consulted with international non-governmental animal health organisations with whom the WOAHA has established a cooperation agreement to also provide feedback on the consolidated Technical Reference Document: HealthforAnimals and the World Veterinary Association.

Endorsement of the final version of the Swine Technical Reference Document was sought from the AMR Working Group and WOAHA hierarchy.

Summary of differences between the antimicrobials listed in the Swine Technical Reference Document and the List of Antimicrobial Agents of Veterinary Importance.

The Subgroup agreed that the following molecules should be included as “used” on the Swine Technical Reference Document, and to keep the reference to swine (SUI) in the WOAHA List: spectinomycin, dihydrostreptomycin, streptomycin, apramycin, gentamicin, kanamycin, neomycin, paromomycin, florfenicol, thiamphenicol, rifaximin, nitarosone, roxarsone, bicozamycin, cefalexin, cefazolin, ceftiofur, ceftriaxone, cefquinome, lincomycin, erythromycin, tulathromycin, josamycin, kitasamycin, mirosamicin, spiramycin, tildipirosin, tilmicosin, tylosin, tylvalosin, avilamycin, benzylpenicillin, benzylpenicillin procaine, amoxicillin, ampicillin, amoxicillin + clavulanic acid, phenoxymethylpenicillin, fosfomicin, tiamulin, valnemulin, bacitracin, enramycin, colistin, flumequine, oxolinic acid, ciprofloxacin, danofloxacin, difloxacin, enrofloxacin, marbofloxacin, norfloxacin, ofloxacin, orbifloxacin, carbadox, phthalylsulfathiazole, sulfachlorpyridazine, sulfadiazine, sulfamethoxazole, sulfadimethoxine, sulfadimidine, sulfadoxine, sulfamerazine, sulfamonomethoxine, sulfapyridine, trimethoprim + sulfonamide, virginamycin, chlortetracycline, doxycycline, oxytetracycline and tetracycline. Sulfamethoxypyridazine was also included as “Used” and moved under “Sulfonamides” class; this antimicrobial agent had been kept previously under “sulfonamides + diaminopyrimidines”. This change will also need to be enacted in the Poultry Technical Reference Document and the WOAHA List. The change in the WOAHA List will be conducted after the main species-specific Technical Reference Documents are finalised.

The Subgroup agreed that the following molecules should be included as “used” on the Swine Technical Reference Document, and to add a reference to swine (SUI) in the WOAHA List: gamithromycin, penethamate hydriodide, polymyxin B, sulfacetamide, sulfaguanidine, sulfanilamide, sulfaquinoxaline, ormetoprim + sulfonamides. The Subgroup supported the change of the listing of “ormetoprim + sulfadimethoxine” to “ormetoprim + sulfonamides” in line with the terminology used for antimicrobial combinations containing trimethoprim. This change will be enacted in the Poultry Technical Reference Document and in the WOAHA List; for the latter, the change will be implemented once all of the main species-specific reference technical documents are finalised.

The Subgroup agreed that the following molecules should not be included on the Swine Technical Reference Document (no change required on the WOAHA List): novobiocin, amikacin, framycetin, tobramycin, rifampicin, cefacetile, cefalonium, cefalotin, cefapirin, cefuroxime, cefoperazone, fusidic acid, lasalocid, maduramicin, monensin, narasin, semduramicin, oleandomycin, carbomycin, benethamine penicillin, hetacillin, ticarcillin, tobicillin, aspoxicillin, pheneticillin, nafcillin, gramicidin, miloxacin, nalidixic acid, sulfafurazole and ormetoprim.

The Subgroup agreed that the following molecules should not be included on the Swine Technical Reference Document, and to remove the reference to swine (SUI) in the WOAHL List: astromycin, salinomycin, pirlimycin, terdecamycin, sedecamycin, mecillinam, ampicillin + sulbactam, aspoxicillin, cloxacillin, dicloxacillin, oxacillin, olaquinox, sulfamethoxydiazine, baquiloprim, trimethoprim and nosiheptide.

The Subgroup agreed to update the nomenclature of the antimicrobial agents in the Swine Technical Reference Document with their Non-Proprietary Names (INNs) in line with international standards. The old names of the antimicrobial agents were kept as synonyms in the list. The names of antimicrobial agents will also be updated in the Poultry Technical Reference Document and in the WOAHL List; the latter will be updated after all the main species-specific technical reference documents are finalised: astromycin (previously listed as fortimycin), cefapirin (cefapirin), mirosamicin (mirosamycin), procaine benzylpenicillin (benzylpenicillin procaine), benzathine benzylpenicillin (benzathine penicillin), polymyxin B (polymyxin B), colistin (polymyxin E), flumequine (flumequin). Furthermore, two sulfonamides that had been previously misnamed were renamed taking into account existing INNs. These will be updated in the Poultry Technical Reference Document and the WOAHL List. The latter will be updated after all the main species-specific technical reference documents are finalised): Sulfamethoxazole (previously named as sulfadimethoxazole) and sulfamethoxydiazine (sulfamethoxine).

The Subgroup discussed several antimicrobial agents proposed for inclusion on the Swine Reference Technical Document that were not previously included on the WOAHL List. The Subgroup agreed to add halquinol (class: halogenated hydroxyquinoline) under the VIA category pending re-evaluation of categorisation at a later stage. The new antimicrobial agent and class will not be added to the WOAHL List until all main species-specific reference technical documents are finalised. The class and the category attributed will be revised at that time.

The Subgroup agreed that toltrazuril and amprolium should not be added to the Swine Technical Reference Document and the WOAHL List as this is an anticoccidial agent without antibacterial activity. The Subgroup agreed not to include any anticoccidial substances without known antimicrobial action in the list as this is not within the remit of the Group. Furthermore, bambamycin and efrotomycin were also not added in the Swine Technical Reference Document as these antimicrobial agents are licensed only for veterinary non-medical use (i.e. growth promotion) in swine.

Thiostrepton has been reclassified as a thiopeptide as this is the most accurate classification for nosiheptide according to current scientific evidence.

It was noted by the Subgroup that Arsenicals have been withdrawn from the market in some countries/regions due to the detection of tissue residues containing inorganic arsenic, a carcinogen.

Criteria for inclusion of pathogens, diseases and antimicrobial classes in the appendices

The list of major bacterial pathogens and diseases affecting swine (Appendix 2) is not intended to be a comprehensive list of all bacterial and microparasitic diseases affecting swine, but to cover those that represent the major health and economical concerns affecting swine reared in commercial production systems. A criterion to include them in the appendix was availability of information for the treatment indications in product labels and country treatment guidelines for swine.

Bacterial and microparasitic pathogens causing diseases in swine for which there is no treatment or treatment are not documented in products labels, were not considered for this document.

The names of the pathogens, in Appendix 2, are listed alphabetically and follow the updated taxonomy. However, where there have been recent taxonomic revisions, older names are also indicated. The names of the diseases caused by the listed pathogens are those most commonly used.

The antimicrobial classes and sub-classes covered in Appendix 3 are only those that include antibiotics which products are authorized for use in swine.

REFERENCES

CUTLER R., GLEESON B., PAGE S., NORRIS J., BROWNING G. (2020). Antimicrobial prescribing guidelines for pigs. 60 pages. <https://www.ava.com.au/siteassets/resources/fighting-antimicrobial-resistance/antimicrobial-prescribing-guidelines-for-pigs.pdf>

The Swedish Veterinary Association (2013). Guidelines for the use of antibiotics in production animals. Cattle, pigs, sheep and goats. 55 pages. <https://www.svf.se/media/vd5ney4l/svfs-riktlinje-antibiotika-till-produktionsdjur-eng-2017.pdf>

Iowa State University. College of Veterinary Medicine. Veterinary Diagnostic and Production Animal Medicine. Swine Disease Manual. <https://vetmed.iastate.edu/vdpam/FSVD/swine/index-diseases/> . Last updated: 2021. Accessed on the 28th December 2021.

OIE (2021). *Terrestrial Animal Health Code*. Paris, France, World Organisation for Animal Health (OIE).

TAYLOR D. J. (1999). Pig Diseases. Seventh edition. Bacterial Diseases:108-252.

ZIMMERMAN J. J., RAMIREZ L.A.K.A., SCHWARTZ K.J., STEVENSON G.W. and ZHANG J. (2019). Diseases of Swine, Eleventh Edition. Hoboken NJ, John Wiley & Sons.

Abbreviations:

The animal species in which antimicrobial agents are used and categories of antimicrobials of veterinary importance are abbreviated as follows:

AVI: Avian
API: Bee
BOV: Bovine
CAN: Canid
CAP: Caprine
CAM: Camel
CRU: Crustaceans
EQU: Equine
FEL: Feline
LEP: Rabbit
OVI: Ovine
PIS: Fish
SUI: Swine
VCIA: Veterinary Critically Important Antimicrobial Agents
VHIA: Veterinary Highly Important Antimicrobial Agents
VIA: Veterinary Important Antimicrobial Agents

Appendices:

Appendix 1a: Members of the Subgroup on Technical Reference Document for Swine.

Appendix 1b: WOA external experts on swine health and non-governmental animal health organisations acting as external reviewers for the Technical Reference Document on Swine.

Appendix 2: List of major pathogens and diseases affecting swine species.

Appendix 3: Antimicrobial classes used in veterinary medicine for swine infections.

ANTIMICROBIAL AGENTS OF VETERINARY IMPORTANCE AUTHORISED FOR USE IN SWINE

Antimicrobial Agents (Class, Sub-class and Substance by International Nonproprietary Name [INN])	Categorisation			Molecules	Species	Authorised for use in swine	Specific comments by class
	VCIA	VHIA	VIA				
AMINOCOUMARIN			x	Novobiocin (vet only)	AVI, CAP, OVI	No	This class is currently only used in animals.
AMINOCYCLITOL	x			Spectinomycin	AVI, BOV, CAN, CAP, EQU, FEL, LEP, OVI, SUI	Yes	
AMINOGLYCOSIDES	x			Dihydrostreptomycin	AVI, BOV, CAN, CAP, EQU, FEL, LEP, OVI, SUI	Yes	Aminoglycosides can be used to treat infections caused by pathogens such as <i>Actinobacillus suis</i> , <i>Bordetella bronchiseptica</i> , <i>Brachyspira hyodysenteriae</i> , <i>E. coli</i> and <i>Lawsonia intracellularis</i> , <i>Pasteurella multocida</i> , <i>Proteus vulgaris</i> , <i>Pseudomonas aeruginosa</i> , <i>Salmonella</i> spp. and <i>Staphylococcus</i> spp.
				Streptomycin	API, AVI, BOV, CAN, CAP, EQU, FEL, LEP, OVI, SUI	Yes	
AMINOGLYCOSIDES + 2 DEOXYSTREPTAMINE	x			Amikacin (synonym: amikacillin, amikacin)	BOV, CAN, EQU, FEL	No	
				Apramycin (vet only)	AVI, BOV, LEP, OVI, SUI	Yes	
				Astromycin (synonyms: fortimycin)	LEP, OVI	No	
				Framycetin	CAN, CAP, FEL, OVI	No	
				Gentamicin	AVI, BOV, CAM, CAN, CAP, EQU, FEL, LEP, OVI, SUI	Yes	
				Kanamycin	AVI, BOV, CAN, EQU, FEL, SUI	Yes	
				Neomycin	API, AVI, BOV, CAN, CAP, CRU, EQU, FEL, LEP, OVI, PIS, SUI	Yes	
				Paromomycin	AVI, BOV, CAN, CAP, FEL, LEP, OVI, SUI	Yes	
				Tobramycin (synonym:	CAN, EQU, FEL	No	

Antimicrobial Agents (Class, Sub-class and Substance by International Nonproprietary Name [INN])	Categorisation			Molecules	Species	Authorised for use in swine	Specific comments by class
	VCIA	VHIA	VIA				
				tobramicin)			
AMPHENICOLS	x			Chloramphenicol	CAN, FEL	No	The wide range of applications and the nature of the diseases treated make amphenicols extremely important for veterinary medicine. Amphenicols can be used to treat respiratory disease caused by <i>Actinobacillus pleuropneumoniae</i> , <i>Bordetella bronchiseptica</i> , <i>Glaesserella parasuis</i> , <i>Mycoplasma hyopneumoniae</i> , <i>Mycoplasma hyorhinis</i> , <i>Pasteurella multocida</i> , <i>Salmonella</i> spp. and <i>Streptococcus suis</i> .
				Florfenicol (vet only)	AVI, BOV, CAN, CAP, CRU, EQU, FEL, LEP, OVI, PIS, SUI	Yes	
				Thiamphenicol	AVI, BOV, CAN, CAP, FEL, OVI, PIS, SUI	Yes	
ANSAMYCINS - RIFAMYCINS		x		Rifampicin (synonym: rifampin)	EQU	No	Ansamycins are used for the topical treatment of skin conditions in swine animals caused by different pathogens such as <i>Staphylococcus</i> spp.
				Rifaximin	BOV, CAN, CAP, EQU, FEL, LEP, OVI, SUI	Yes	
ARSENICALS			x	Nitarsonsone (vet only)	AVI, SUI	Yes	Arsenicals Are used to treat <i>Brachyspira hyodysenteriae</i> . Note: Arsenicals have been withdrawn from the market in some countries/regions due to the detection of tissue residues containing inorganic arsenic, a carcinogen. This class is currently only used in animals.
				Roxarsone (vet only)	AVI, SUI	Yes	
BICYCLOMYCIN			x	Bicozamycin (synonym: bicyclomycin)	SUI	Yes	Bicyclomycin is used to treat <i>E. coli</i> and <i>Salmonella</i> spp. Infections.
CEPHALOSPORINS		x					
Cephalosporin 1st Generation				Cefacetrile (synonyms: cephacetrile, cefacetril, cephacetril)	BOV	No	First and second generation cephalosporins are used in the treatment of septicaemias, respiratory infections, and mastitis in food-producing animals.
				Cefalexin (synonyms: cephalexin, cephacillin, cephalaxine, cefalexine)	AVI, BOV, CAN, CAP, EQU, FEL, OVI, SUI	yes	
				Cefalonium (vet only) (synonyms: cephalonium, cefalonum)	BOV, CAN, CAP, OVI	No	

Antimicrobial Agents (Class, Sub-class and Substance by International Nonproprietary Name [INN])	Categorisation			Molecules	Species	Authorised for use in swine	Specific comments by class
	VCIA	VHIA	VIA				
				Cefalotin	BOV, CAN, EQU	No	
				Cefapirin (synonyms: cephapirin, cefapyrin)	BOV	No	
				Cefazolin (synonyms: cephazolin, cephaloline, cephazolidin)	BOV, CAP, OVI, SUI	Yes	
Cephalosporin 2nd Generation				Cefuroxime	BOV	No	
Cephalosporin 3rd Generation	x			Cefixime	CAN, FEL	No	Third and fourth generation cephalosporins are critically important for human health and subject to specific recommendations in the WOAHP List of Antimicrobial Agents of Veterinary Importance. Their use in animals should only occur when the pathogen is resistant to the first choice antimicrobial; its use should be supported by antimicrobial susceptibility testing whenever possible. Extra-label/off label use should be limited and reserved for instances where no alternatives are available and in agreement with national legislation. Third and fourth generation cephalosporins are indicated for treat several diseases caused by Gram-negative and Gram-positive bacteria such as <i>Actinobacillus pleuropneumoniae</i> , <i>E. coli</i> , <i>Glaesserella parasuis</i> , <i>Pasteurella multocida</i> , <i>Salmonella</i> spp., <i>Staphylococcus</i> spp., and <i>Streptococcus</i> spp.
				Cefoperazone	BOV, CAP, OVI	No	
				Cefovecin (vet only)	CAN, FEL	No	
				Cefpodoxime	CAN	No	
				Ceftiofur (vet only)	AVI, BOV, CAN, CAP, EQU, LEP, OVI, SUI	Yes	
				Ceftriaxone	BOV, CAN, OVI, SUI	Yes	
Cephalosporin 4th Generation				Cefquinome (vet only)	BOV, CAP, EQU, LEP, OVI, SUI	Yes	
FUSIDANE			x	Fusidic acid	CAN, EQU, FEL	No	
IONOPHORES		x		Lasalocid (vet only)	AVI, BOV, LEP, OVI	No	Ionophores are essential for animal health because they are used to control intestinal parasitic coccidiosis where there are few or no alternatives available, and also have antimicrobial properties. This class is currently only used in animals.
				Maduramicin (vet only)	AVI	No	
				Monensin (vet only)	API, AVI, BOV, CAP	No	
				Narasin (vet only)	AVI	No	
				Salinomycin (vet only)	AVI, LEP	No	
				Semduramicin (vet only)	AVI	No	

Antimicrobial Agents (Class, Sub-class and Substance by International Nonproprietary Name [INN])	Categorisation			Molecules	Species	Authorised for use in swine	Specific comments by class
	VCIA	VHIA	VIA				
LINCOSAMIDES		x		Clindamycin	CAN, FEL	No	Lincosamides are very important for the treatment of infections caused by Gram positive and anaerobic bacteria such as <i>Bacteroides</i> spp., <i>Brachyspira hyodysenteriae</i> , <i>Erysipelothrix rhusiopathiae</i> , <i>Fusobacterium</i> spp., <i>Lawsonia intracellularis</i> and <i>Mycoplasma</i> spp., <i>Staphylococcus</i> spp. and <i>Streptococcus</i> spp.
				Lincomycin	API, AVI, BOV, CAN, CAP, FEL, OVI, PIS, SUI	Yes	
				Pirlimycin (vet only)	BOV	No	
MACROLIDES	x						
Macrolides 14-membered ring				Erythromycin	API, AVI, BOV,CAN, CAP, EQU, FEL, LEP, OVI, PIS, SUI	Yes	The wide range of applications and the nature of the diseases treated make macrolides extremely important for veterinary medicine.
				Oleandomycin		No	
Macrolides 15-membered ring				Azithromycin	CAN	No	Macrolides are indicated to treat respiratory diseases caused by <i>Actinobacillus pleuropneumoniae</i> , <i>Bordetella bronchiseptica</i> , <i>Brachyspira hyodysenteriae</i> , <i>Glaesserella parasuis</i> , <i>Lawsonia intracellularis</i> , <i>Mycoplasma hyopneumoniae</i> , <i>Pasteurella multocida</i> , and <i>Streptococcus</i> spp.
				Gamithromycin (vet only)	BOV, SUI	Yes	
				Tulathromycin (vet only)	BOV, SUI	Yes	
Macrolides 16-membered ring				Carbomycin	AVI	No	
				Josamycin	SUI	Yes	
				Kitasamycin (vet only)	AVI, PIS, SUI	Yes	
				Mirosamicin (synonyms: mirosamycin, miporamycin)	API, AVI, SUI	Yes	
				Spiramycin	AVI, BOV, CAP, EQU, LEP, OVI, SUI	Yes	
				Tildipirosin (vet only)	BOV, SUI	Yes	
				Tilmicosin (vet only)	AVI, BOV, CAP, LEP, OVI, PIS, SUI	Yes	
				Tylosin (vet only)	API, AVI, BOV, CAP, LEP, OVI, SUI	Yes	
			Tylvalosin (vet only)	AVI, SUI	Yes		
Macrolides 17-membered ring				Sedecamycin (synonym:		No	

Antimicrobial Agents (Class, Sub-class and Substance by International Nonproprietary Name [INN])	Categorisation			Molecules	Species	Authorised for use in swine	Specific comments by class
	VCIA	VHIA	VIA				
				lankacidin A)			
				Terdecamycin		No	
ORTHOSOMYCINS			x	Avilamycin (vet only)	AVI, LEP, SUI	Yes	Avilamycin is used in the treatment of <i>E. coli</i> infections This class is currently only used in animals.
PENICILLINS	x						
Natural penicillins (including esters and salts)				Benethamine penicillin		No	The wide range of applications and the nature of the diseases treated make penicillins extremely important for swine. Agents within this class are used in swine to treat infections caused by pathogens such as <i>Actinobacillus pleuropneumoniae</i> , <i>Actinobacillus suis</i> , <i>Actinobaculus suis</i> , <i>Brachyspira hyodysenteriae</i> , <i>Erysipelothrix rhusiopathiae</i> , <i>E. coli</i> , <i>Clostridium perfringens</i> , <i>Glaesserella parasuis</i> , <i>Pasteurella multocida</i> , <i>Staphylococcus</i> spp., <i>Streptococcus</i> spp.
				Benzylpenicillin (synonyms: penicillin G, benzylpenicillin G, benzopenicillin, benzyl Penicillin)	AVI, BOV, CAM, CAN, CAP, EQU, FEL, LEP, OVI, SUI	Yes	
				Procaine benzylpenicillin (synonyms: benzylpenicillin procaine, procaine G penicillin) Benzathine benzylpenicillin (synonyms: benzathine penicillin, benzathine penicillin G)	BOV, CAM, CAN, CAP, EQU, FEL, OVI, SUI	Yes	
				Penethamate hydriodide (vet only)	BOV, CAN, SUI	Yes	
				Tobicillin		No	
Amidinopenicillins				Mecillinam (synonyms: amdinocillin, hexacillin, penicillin HX)		No	
Aminopenicillins				Amoxicillin (synonym: amoxycillin)	AVI, BOV, CAN, CAP, EQU, FEL, OVI, PIS, SUI	Yes	
				Ampicillin	AVI, BOV, CAN, CAP, EQU, FEL, OVI, PIS, SUI	Yes	

Antimicrobial Agents (Class, Sub-class and Substance by International Nonproprietary Name [INN])	Categorisation			Molecules	Species	Authorised for use in swine	Specific comments by class
	VCIA	VHIA	VIA				
				Hetacillin (synonym: phenazacillin)	BOV	No	
Aminopenicillin plus betalactamase inhibitor				Amoxicillin + clavulanic acid	AVI, BOV, CAN, CAP, EQU, FEL, OVI, SUI	Yes	
Carboxypenicillins				Ampicillin + sulbactam	BOV	No	
Phenoxypenicillins				Ticarcillin	EQU	No	
				Pheneticillin (synonyms: phenethicillin, penicillin B)	EQU	No	
				Phenoxymethylpenicillin (synonyms: penicillin V, pen V, penicillin phenoxymethyl, phenoxymethyl penicillin, beromycin, oraxillin)	AVI, CAN, SUI	Yes	
Antistaphylococcal penicillins				Cloxacillin (synonym: methocillin S)	BOV, CAN, CAP, EQU, FEL, OVI	No	
				Dicloxacillin (synonym: dicloxacycline)	BOV, CAP, EQU, OVI	No	
				Nafcillin (synonym: naphcillin)	CAP, OVI	No	
				Oxacillin (synonyms: oxazocillin, MPI-penicillin)	BOV, CAP, EQU, OVI	No	
Antipseudomonal penicillins				Aspoxicillin		No	
PHOSPHONIC ACID DERIVATIVES		x		Fosfomycin (synonyms: phosphomycin, phosphonomycin)	AVI, BOV, PIS, SUI	Yes	Phosphonic acid derivatives are critically important for human health and subject to specific recommendations in the WOAHA List of Antimicrobial Agents of Veterinary Importance. Their use in animals should only occur when the pathogen is resistant to the first choice antimicrobial; its use should be supported by antimicrobial susceptibility testing whenever possible. Extra-label/off label use should be limited and reserved for instances where no alternatives are available and in

Antimicrobial Agents (Class, Sub-class and Substance by International Nonproprietary Name [INN])	Categorisation			Molecules	Species	Authorised for use in swine	Specific comments by class
	VCIA	VHIA	VIA				
							agreement with national legislation. Phosphonic Acid Derivatives are indicated for control and treatment of swine diseases caused by pathogens such as: <i>Actinobacillus pleuropneumoniae</i> , <i>E. coli</i> , <i>Pasteurella multocida</i> , <i>Salmonella</i> spp., <i>Staphylococcus aureus</i> and <i>Streptococcus</i> spp.
PLEUROMUTILINS		x		Tiamulin (vet only) (synonym: thiamutilin)	AVI, CAP, LEP, OVI, PIS, SUI	Yes	Pleuromutilins are used in treatment of systemic and enteric diseases in swine caused by <i>Actinobacillus pleuropneumoniae</i> , <i>Brachyspira hyodysenteriae</i> , <i>B. pilosicoli</i> , <i>Glaesserella parasuis</i> , <i>Lawsonia intracellularis</i> , <i>Mycoplasma hyopneumoniae</i> , <i>Pasteurella multocida</i> and <i>Streptococcus suis</i> .
				Valnemulin (vet only)	SUI	Yes	
POLYPEPTIDES							
Cyclic polypeptides		x		Bacitracin	AVI, BOV, CAN, FEL, LEP, OVI, SUI	Yes	Polypeptides have a specific action on the enteric microbiota, especially <i>Brachyspira hyodysenteriae</i> , <i>Clostridium perfringens</i> , <i>Staphylococcus</i> spp. and <i>Streptococcus</i> spp.
				Enramycin	AVI, SUI	Yes	
				Gramicidin	EQU	No	
Polymyxins				Polymyxin B (synonym: polymixin B)	CAN, CAP, EQU, FEL, LEP, OVI, SUI	Yes	Polymyxins are indicated mainly for the treatment and control of diarrhea caused by gram-negative bacteria such as: <i>E. coli</i> and <i>Salmonella</i> spp. and in case of mixed infections. Colistin is critically important for human health and subject to specific recommendations in the WOA List of Antimicrobial Agents of Veterinary Importance. Its use in animals should only occur when the pathogen is resistant to the first choice antimicrobial; its use should be supported by antimicrobial susceptibility testing whenever possible. Extra-label/off label use should be limited and reserved for instances where no alternatives are available and in agreement with national legislation.
				Colistin (synonym: polymyxin E)	AVI, BOV, CAN, CAP, EQU, FEL, LEP, OVI, SUI	Yes	
QUINOLONES							
Quinolones 1st Generation		x		Flumequine (synonym:	AVI, BOV, CAN, CAP, EQU,	Yes	Quinolones of the 1st Generation are used in the

Antimicrobial Agents (Class, Sub-class and Substance by International Nonproprietary Name [INN])	Categorisation			Molecules	Species	Authorised for use in swine	Specific comments by class
	VCIA	VHIA	VIA				
				flumequin)	FEL, LEP, OVI, PIS, SUI		treatment of respiratory infections caused by <i>Bordetella</i> spp., <i>E. coli</i> and <i>Pasteurella</i> spp.
				Miloxacin		No	
				Nalidixic acid (synonyms: nalixidate, nalidixinic acid, nalidic acid)		No	
				Oxolinic acid	AVI, BOV, LEP, OVI, PIS, SUI	Yes	
Quinolones 2nd Generation (Fluoroquinolones)	x			Ciprofloxacin	AVI, BOV, PIS, SUI	Yes	Fluoroquinolones are critically important for human health and subject to specific recommendations in the WOAHA List of Antimicrobial Agents of Veterinary Importance. Its use in animals should only occur when the pathogen is resistant to the first choice antimicrobial; its use should be supported by antimicrobial susceptibility testing whenever possible. Extra-label/off label use should be limited and reserved for instances where no alternatives are available and in agreement with national legislation.
				Danofloxacin (vet only)	BOV, CAP, LEP, OVI, SUI	Yes	
				Difloxacin	AVI, LEP, SUI	Yes	
				Enrofloxacin (vet only)	AVI, BOV, CAN, CAP, CRU, EQU, FEL, LEP, OVI, PIS, SUI	Yes	
				Ibafloxacin	CAN, FEL	No	
				Levofloxacin	CAN	No	
				Marbofloxacin (vet only)	BOV, CAN, EQU, FEL, LEP, SUI	Yes	Quinolones 2nd Generation (Fluoroquinolones) are used to treat infections caused by pathogens such as <i>Actinobacillus pleuropneumoniae</i> , <i>Actinobacillus suis</i> , <i>Bordetella bronchiseptica</i> , <i>Erysipelothrix rhusiopathiae</i> , <i>E. coli</i> , <i>Glaesserella parasuis</i> , <i>Klebsiella</i> spp., <i>Mycoplasma hyopneumoniae</i> , <i>Pasteurella multocida</i> , <i>Proteus mirabilis</i> , <i>Salmonella</i> spp., <i>Staphylococcus</i> spp., <i>Streptococcus dysgalactiae</i> , <i>Streptococcus pneumoniae</i> , <i>Streptococcus suis</i> and <i>Trueperella pyogenes</i> .
				Norfloxacin	AVI, BOV, CAN, CAP, FEL, LEP, OVI, SUI	Yes	
				Ofloxacin	AVI, CAN, FEL, SUI	Yes	
				Orbifloxacin (vet only)	BOV, CAN, FEL, SUI	Yes	
				Pradofloxacin (vet only)	BOV, CAN, FEL	No	
				Sarafloxacin		No	
QUINOXALINES			x	Carbadox (vet only)	SUI	Yes	Carbadox is used for the treatment of gastrointestinal infections caused by <i>Brachyspira</i> spp., <i>Lawsonia intracellularis</i> and <i>Salmonella</i> spp.
				Olaquinox (vet only) (synonym: olachinox)		No	

Antimicrobial Agents (Class, Sub-class and Substance by International Nonproprietary Name [INN])	Categorisation			Molecules	Species	Authorised for use in swine	Specific comments by class
	VCIA	VHIA	VIA				
							This class is currently only used in animals. Note: Carbadox has been withdrawn from the market in some countries/regions due to the detection of carcinogenic tissue residues.
SULFONAMIDES	x			Phthalylsulfathiazole (vet only) (synonyms: sulfathalidine, phthalazol, phthalylsulphathiazole, phthalylsulfonazole)	CAN, FEL, SUI	Yes	The wide range of applications and the nature of the diseases treated make sulfonamides (sulfas) extremely important for swine. Sulfonamides are used to treat infections caused by pathogens such as <i>Actinobacillus pleuropneumoniae</i>, coccidia, <i>E. coli</i>, <i>Glaesserella parasuis</i>, <i>Pasteurella multocida</i>, <i>Salmonella</i> spp., <i>Staphylococcus</i> spp., <i>Streptococcus</i> spp. Often used in combination with trimethoprim.
				Sulfacetamide (synonyms: sulphacetamide, acetosulfamine, acetosulfamin, N-acetylsulfanilamide)	AVI, BOV, CAN, FEL, OVI, SUI	Yes	
				Sulfachlorpyridazine (synonym: sulfachloropyridazine)	AVI, BOV, SUI	Yes	
				Sulfadiazine (synonyms: sulphadiazine, sulfapyrimidine, sulfadiazin, sulfazine, sulfadiazene)	AVI, BOV, CAN, CAP, FEL, OVI, PIS, SUI	Yes	
				Sulfamethoxazole (synonyms: sulfadimethoxazole, sulphamethoxazole, sulfisomezole)	AVI, BOV, CAN, FEL, SUI	Yes	
				Sulfadimethoxine (synonyms: sulphadimethoxine, sulfadimethoxin, sulfadimethoxydiazine)	AVI, BOV, CAN, CAP, EQU, FEL, LEP, OVI, PIS, SUI	Yes	
				Sulfadimidine (synonyms: sulfamethazine, sulfadimethyldiazine,	AVI, BOV, CAN, CAP, EQU, FEL, LEP, OVI, SUI	Yes	

Antimicrobial Agents (Class, Sub-class and Substance by International Nonproprietary Name [INN])	Categorisation			Molecules	Species	Authorised for use in swine	Specific comments by class
	VCIA	VHIA	VIA				
				sulfamezathine, sulphamethazine, sulfadimerazine)			
				Sulfadoxine (synonyms: sulphadoxine, sulforthomidine, sulphormethoxine, sulfadoxin)	AVI, BOV, CAN, EQU, FEL, OVI, SUI	Yes	
				Sulfafurazole (synonyms: sulfisoxazole, sulphafurazole, sulfisoxazol, sulfafurazol)	CAN, PIS	No	
				Sulfaguanidine (synonyms: sulfaguanidin, sulphaguanidine, sulfanilguanidine, sulfoguanidine)	AVI, BOV, CAN, CAP, FEL, OVI, SUI	Yes	
				Sulfamerazine (synonyms: sulphamerazine, sulfamerazin, sulfamethyldiazine)	AVI, BOV, CAN, CAP, EQU, FEL, LEP, OVI, PIS, SUI	Yes	
				Sulfamethoxydiazine (synonyms: sulfamethoxine, sulfameter, sulfamethoxydiazine, sulfamethoxypyrimidine)	AVI	No	
				Sulfamonomethoxine (synonyms: sulfamonomethoxin, sulfamonmethoxine)	AVI, BOV, CAN, FEL, PIS, SUI	Yes	
				Sulfanilamide (synonyms: sulphanilamide, sulfamine, sulfonilamide)	BOV, CAN, CAP, FEL, OVI, SUI	Yes	
				Sulfapyridine (synonym:	BOV, CAN, FEL, SUI	Yes	

Antimicrobial Agents (Class, Sub-class and Substance by International Nonproprietary Name [INN])	Categorisation			Molecules	Species	Authorised for use in swine	Specific comments by class
	VCIA	VHIA	VIA				
Sulfonamides + diaminopyrimidines				sulphapyridine)			
				Sulfaquinoxaline (synonyms: sulfabenzpyrazine, sulphaquinoxaline)	AVI, BOV, CAP, LEP, OVI, SUI	Yes	
				Sulfamethoxypyridazine (synonyms: sulphamethoxypyridazine, sulfapyridazine, sulfametoxipiridazine)	AVI, BOV, CAN, EQU, FEL, SUI	Yes	
				Ormetoprim (synonyms: ormethoprim, ormetorprim) + sulfonamide	AVI, BOV, PIS, SUI	Yes	
				Trimethoprim (synonym: trimetoprim) + sulfonamide	AVI, BOV, CAN, CAP, EQU, FEL, LEP, OVI, PIS, SUI	Yes	
				Baquiloprim		No	
				Ormetoprim (synonyms: ormethoprim, ormetorprim)	AVI	No	
DIAMINOPYRIMIDINES				Trimethoprim (synonym: trimetoprim)	AVI, BOV, CAP, EQU, LEP, OVI	No	
STREPTOGRAMINS			x	Virginiamycin (vet only)	AVI, BOV, OVI, SUI	Yes	Virginiamycin important antimicrobial for the treatment of infections caused by <i>Brachyspira hyodysenteriae</i> and <i>Lawsonia intracellularis</i> .
TETRACYCLINES	x			Chlortetracycline	AVI, BOV, CAN, CAP, EQU, FEL, LEP, OVI, PIS, SUI	Yes	The wide range of applications and the nature of the diseases treated make tetracyclines extremely important for swine. This class alone or in combination is critically important in the treatment of a wide range of diseases caused by <i>Actinobacillus pleuropneumoniae</i> , <i>Bordetella bronchiseptica</i> , <i>Erysipelothrix rhusiopathiae</i> , <i>E. coli</i> , <i>Klebsiella</i> spp., <i>Lawsonia intracellularis</i> , <i>Mycoplasma hyopneumoniae</i> , <i>Pasteurella multocida</i> , <i>Salmonella</i> spp., <i>Staphylococcus aureus</i> and <i>Streptococcus suis</i> .
				Doxycycline (synonyms: doxytetracycline, doxycyclin)	AVI, BOV, CAM, CAN, CAP, EQU, FEL, LEP, OVI, PIS, SUI	Yes	
				Oxytetracycline (synonyms: oxytetracycline, oxytetracyclin, oxitetraacyclin) oxytetracyne)	API, AVI, BOV, CAM, CAN, CAP, CRU, EQU, FEL, LEP, OVI, PIS, SUI	Yes	
				Tetracycline (synonym: tetracyclin)	API, AVI, BOV, CAM, CAN, CAP, EQU, FEL, LEP, OVI,	Yes	

Antimicrobial Agents (Class, Sub-class and Substance by International Nonproprietary Name [INN])	Categorisation			Molecules	Species	Authorised for use in swine	Specific comments by class
	VCIA	VHIA	VIA				
					PIS, SUI		
THIOPEPTIDES			x	Nosiheptide	BOV	No	
				Thiostrepton	CAN, FEL	No	
HALOGENATED HYDROXYQUINOLINES			x	Halquinol	SUI	Yes	Treatment of diarrhoea caused by <i>E. coli</i> and <i>Salmonella</i> spp.
PSEUDOMONIC ACID				Mupirocin	CAN, FEL	No	
NITROMIDAZOLE				Metronidazole	CAN, FEL	No	
				Ornidazole	CAN	No	
				Tinidazole	CAN, FEL	No	

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LIST OF MAJOR PATHOGENS AND DISEASES AFFECTING SWINE SPECIES

Pathogens	Examples of diseases
Bacteria	
<i>Actinobacillus pleuropneumoniae</i>	Pleuropneumonia
<i>Actinobacillus suis</i>	Pneumonia, septicaemia
<i>Actinobaculum suis</i> (<i>Eubacterium suis</i>)	Cystitis, pyelonephritis
<i>Bordetella bronchiseptica</i>	Nonprogressive atrophic rhinitis, pneumonia
<i>Brachyspira</i> (<i>Serpulina</i> , <i>Treponema</i>) <i>hyodysenteriae</i> , <i>B. hampsonii</i> , <i>B. suanatina</i>	Swine dysentery
<i>Brachyspira pilosicoli</i>	Porcine colonic spirochetosis (PCS) or intestinal spirochetosis
<i>Clostridium perfringens</i> type A	Mild enteritis, sometimes pseudomembranous chronic enteritis, significant stunting of growth of affected pigs, sudden death in sows.
<i>Clostridium perfringens</i> type C	Neonatal haemorrhagic, necrotic enteritis type C
<i>E. coli</i>	Colibacillosis, edema disease, cystitis, enteritis, mastitis, lactation failure, neonatal septicemia
<i>Erysipelothrix rhusiopathiae</i>	Erysipelas: Septicaemia, arthritis, endocarditis
<i>Glaesserella</i> (<i>Haemophilus</i>) <i>parasuis</i>	Glässer's Disease, fibrinous polyserositis and arthritis
<i>Klebsiella pneumoniae</i>	Septicaemia in piglets, severe coliform mastitis
<i>Lawsonia intracellularis</i>	Acute proliferative haemorrhagic enteropathy (PHE), chronic porcine proliferative enteropathy (PPE)
<i>Mycoplasma hyopneumoniae</i>	Enzootic pneumonia
<i>Mycoplasma hyorhinis</i>	Polyserositis, arthritis
<i>Mycoplasma hyosynoviae</i>	Arthritis
<i>Mycoplasma</i> (<i>Hemoplasma</i> , <i>Eperythrozoon</i>) <i>suis</i>	Anaemia, icterus, infertility, lactation failure, ear and tail necrosis, pericarditis, unthriftiness
<i>Pasteurella multocida</i>	Progressive atrophic rhinitis, pneumonia, septicaemia
<i>Rhodococcus equi</i>	Granulomatous bronchopneumonia, lymphadenitis
<i>Salmonella enterica</i>	Enterocolitis, meningitis, pneumonia, septicaemia
<i>Staphylococcus aureus</i>	Neonatal septicaemia, subcutaneous abscesses, arthritis, mastitis, vaginitis, metritis and agalactia.
<i>Staphylococcus hyicus</i>	Exudative epidermitis
<i>Streptococcus suis</i>	Septicaemia, meningitis, arthritis, endocarditis, polyserositis
<i>Streptococcus</i> spp.	Navel infection, polyarthritis, pneumonia, polyserositis, impetigo-like infectious dermatitis, upper respiratory disease, pleuritis, pericarditis, mastitis, lymphadenitis, infertility syndromes.
<i>Trueperella</i> (<i>Corynebacterium</i> , <i>Actinomyces</i> , <i>Arcanobacterium</i>) <i>pyogenes</i>	Vertebral osteomyelitis, arthritis, pneumonia, endocarditis, mastitis, subcutaneous and deep tissue abscesses
<i>Yersinia</i> spp.	Enterocolitis
Coccidia	
<i>Eimeria deblickei</i> , <i>E. neodeblickei</i> , <i>E. scabra</i> , and <i>E. spinosa</i>	Diarrhoea, unthriftiness in piglets
<i>Cystoisospora</i> (<i>Isospora</i>) <i>suis</i>	Diarrhoea in piglets

ANTIMICROBIAL CLASSES USED IN VETERINARY MEDICINE FOR SWINE INFECTIONS

	Actinobacillus pleuropneumoniae infection	A. suis and A. equuli infection	Actinobaculum suis infection	Bordetella bronchiseptica infection	Brachyspira hyodysenteriae, B. hampsonii, B. suanatina infection	B. pilosicoli infection	Coccidia infections	Clostridium perfringens infections	Erysipelothrix spp. infection	E. coli infection	Glaesserella parasuis infection	Klebsiella pneumoniae infection	Lawsonia intracellularis infection	Mycoplasma hyopneumoniae infection	M. hyorhinis and M. hyosynoviae infection	Pasteurella multocida infection	Rhodococcus equi infection	Salmonella spp. infection	Staphylococcus spp.	Streptococcus spp. infections	Trueperella pyogenes infection	Yersinia spp. infection
AMINOCYCLITOL				X					X	X		X	X			X		X				
AMINOGLYCOSIDES				X					X			X						X	X			
AMINOGLYCOSIDES + 2 DEOXYSTREPTAMINE				X	X	X			X			X						X	X			
AMPHENICOLS	X			X							X			X	X	X		X		X		
ANSAMYCINS - RIFAMYCINS																			X			
ARSENICALS					X																	
BICYCLOMYCIN									X									X				
CEPHALOSPORINS	X								X	X						X	X	X	X	X		
LINCOSAMIDES					X			X				X	X	X					X	X		
MACROLIDES	X			X	X						X		X	X		X	X			X		
ORTHOSOMYCINS									X													
PENICILLINS	X	X			X			X	X	X	X					X	X		X	X	X	
PHOSPHONIC ACID DERIVATIVES	X								X							X			X	X		
PLEUROMUTILINS	X				X	X					X		X	X		X				X		
POLYMYXINS									X									X				
POLYPEPTIDES (OTHER THAN POLYMYXINS)					X			X											X			
QUINOLONES	X	X		X					X	X	X	X		X		X	X	X	X	X	X	X
QUINOXALINES					X	X							X					X				
SULFONAMIDES	X						X		X	X						X		X	X	X		
SULFONAMIDES + DIAMINOPYRIMIDINES	X						X		X	X						X		X	X	X		
STREPTOGRAMINS					X								X									
TETRACYCLINES	X			X					X	X		X	X	X		X		X	X	X		
QUINOLINES									X									X				